

Backup, Restore, and Database Cloning

Managing databases involves two critical skills: **Safety** (Backups) and **Cloning** (Copying). In PostgreSQL, these tasks are primarily handled through the command line using utilities like `pg_dump` and `psql`.

1. Database Backup (`pg_dump`)

A backup creates a text file containing the SQL commands necessary to reconstruct the database to the exact state it was in at the time of the backup.

- **Command:** `pg_dump -U username -d dbname > backup_file.sql`
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- **Key Flags:**
 - `-U`: Specifies the database user (e.g., `postgres`).
 - `-d`: The name of the database you want to back up.
 - `>`: Redirects the output into a file.

2. Database Restore (`psql`)

Restoring involves running the SQL script generated by the backup against a fresh database.

- **Step 1:** Create an empty database: `CREATE DATABASE new_db;`
 - **Step 2:** Run the restore command:
 - `psql -U username -d new_db < backup_file.sql`
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3. How to Copy a Database

Sometimes you need to duplicate a database on the same server (e.g., creating a "test" version of a "production" database). PostgreSQL makes this easy with the `TEMPLATE` clause.

SQL Syntax:

```
CREATE DATABASE destination_db
WITH TEMPLATE source_db;
```

Note: You must disconnect all users from the `source_db` before running this command, or the copy will fail.

Tasks

Complete these tasks to master the workflow. Assume your username is `postgres`.

Task 1: The Safety Net

1. Create a database named `school_db`.

2. Create a table named `students` with columns for `id` and `name`.
3. Insert two rows of data.
4. **Action:** Run a command in your terminal to back up `school_db` to a file named `school_backup.sql`.

Task 2: The Time Machine

1. Drop your `school_db` database (`DROP DATABASE school_db;`).
2. Create a brand new, empty database called `school_recovered`.
3. **Action:** Restore the data from `school_backup.sql` into `school_recovered`. Verify the students are still there.

Task 3: The Sandbox

1. You want to experiment with new features without breaking `school_recovered`.
2. **Action:** Use the `TEMPLATE` command to create a copy of `school_recovered` named `school_test`.

Troubleshooting Tips

- **Role issues:** If you get a "Permission Denied" error, ensure your user has `SUPERUSER` or `CREATEDB` privileges.
- **Pathing:** On Windows, you may need to navigate to the PostgreSQL `bin` folder (e.g., `C:\Program Files\PostgreSQL\16\bin`) in your command prompt to run `pg_dump`.
- **Active Connections:** If you can't copy a database because it's "being accessed by other users," run this query to kick everyone off:

- ```
SELECT pg_terminate_backend(pg_stat_activity.pid)
FROM pg_stat_activity
WHERE pg_stat_activity.datname = 'source_db_name'
AND pid <> pg_backend_pid();
```